

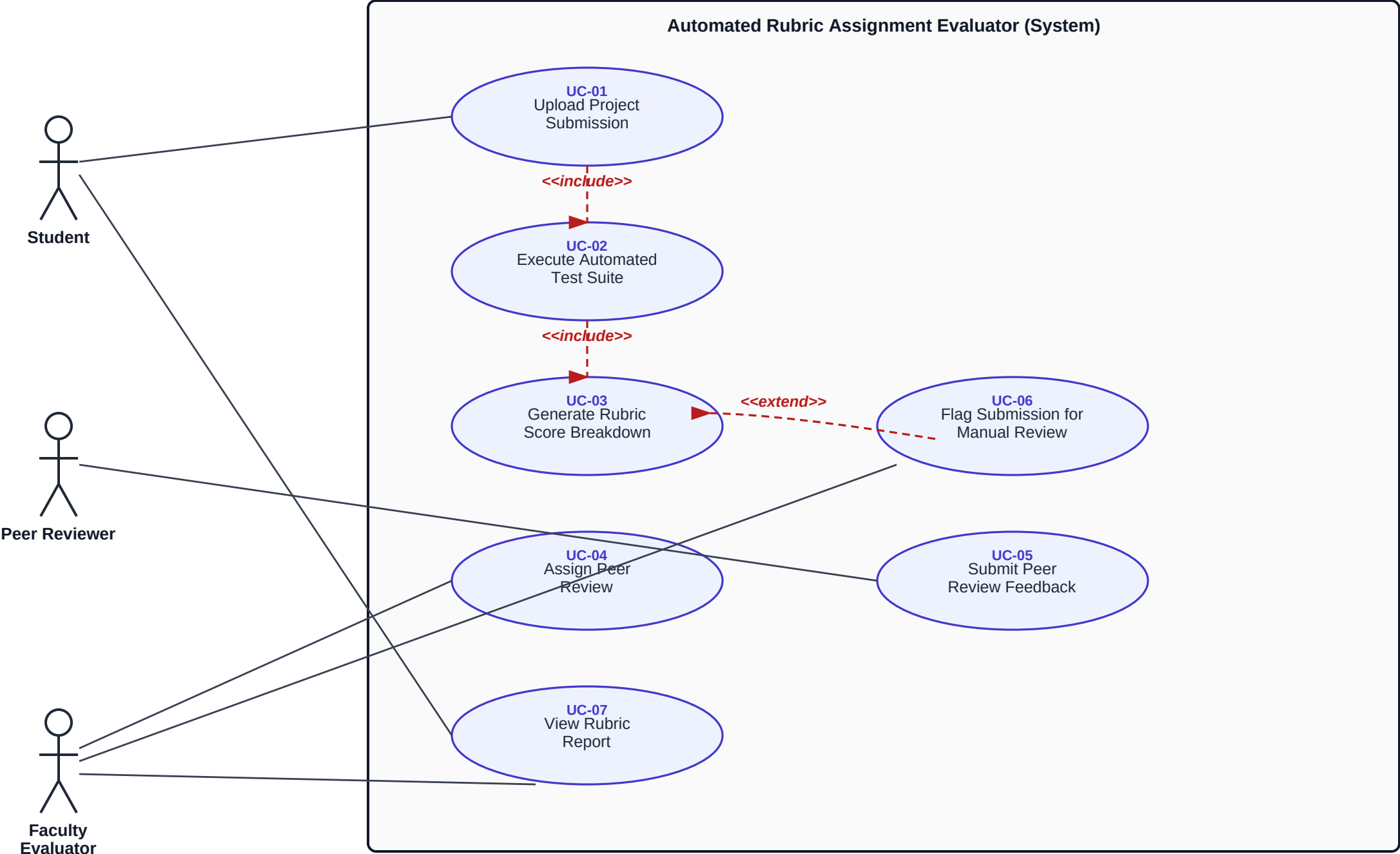
Lab 1 — Requirements Table

Problem Statement #02 — Automated Rubric Assignment Evaluator

Actors: Student, Faculty Evaluator, Peer Reviewer

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall automatically queue uploaded student project zip files, execute pre-configured unit test suites, and generate an itemized rubric score breakdown.	High	Pass: Rubric breakdown is rendered with test results within 10 seconds of submission. Fail: File queue stalls or invalid rubric tally.	Automates the core grading pipeline so evaluators are not manually running tests per submission. (Given)
FR-002	Functional	The system shall automatically assign each graded submission to exactly two peer reviewers, excluding the submission's own author.	Medium	Pass: Every submission has exactly 2 reviewers assigned within 5 seconds of scoring completion, with no self-assignment. Fail: A submission has fewer than 2 reviewers or is assigned to its own author.	Distributes peer-review workload evenly and prevents biased self-review.
FR-003	Functional	The system shall allow a faculty evaluator to manually override or adjust an automatically generated rubric score, with the change logged.	High	Pass: Override is saved and an audit entry records original score, new score, evaluator ID, and timestamp. Fail: Override is not persisted or no audit entry is created.	Automated scoring can misjudge edge cases; faculty need final authority and a traceable record.
FR-004	Functional	The system shall notify a student via email and dashboard once their submission's test execution and rubric scoring are complete.	Medium	Pass: Notification is sent within 30 seconds of scoring completion. Fail: Notification is delayed beyond 30 seconds or not sent.	Keeps students informed of grading status without needing to poll the portal.
FR-005	Functional	The system shall flag any submission whose automated rubric score falls within a low-confidence variance threshold for mandatory manual faculty review.	Medium	Pass: All submissions above the configured variance threshold appear in the faculty 'Needs Review' queue. Fail: A flagged submission is missing from the queue.	Ensures ambiguous or borderline automated grades receive human verification before being finalized.
NFR-001	Nonfunctional	The system shall handle up to 100 concurrent project submissions without dropping queue items or exceeding 1.5 GB memory footprint.	High	Pass: Benchmarking tests confirm target latency and stable memory use under simulated peak load.	Ensures the platform stays responsive during deadline-driven submission spikes. (Given)
NFR-002	Nonfunctional	The system shall store all submitted code, test results, and scores in an encrypted data store, accessible only to the submitting student and authorized faculty.	High	Pass: Storage audit confirms encryption at rest; access-control tests show unauthorized read/write attempts are denied. Fail: Data is stored unencrypted or an unauthorized access attempt succeeds.	Protects student academic records and satisfies institutional data-privacy requirements.

Use-Case Diagram: Automated Rubric Assignment Evaluator



Lab 1 — Use-Case Flow

Problem Statement #02 — Automated Rubric Assignment Evaluator

Use Case: UC-03 — Generate Rubric Score Breakdown

Primary Actor: Student (triggers via submission) | Supporting Actor: Faculty Evaluator

Preconditions

- Student has an active account and a valid assignment is open for submission.
- The pre-configured unit test suite and rubric weighting for the assignment exist in the system.

Postconditions

- An itemized rubric score breakdown is generated and stored against the submission.
- The student and faculty evaluator can view the report; peer-review assignment (UC-04) becomes eligible to run.

Main Success Scenario

1. Student selects "Upload Submission" and uploads a zipped project file (includes UC-01).
2. System validates the file structure and confirms it meets the assignment's submission format.
3. System automatically queues the submission and executes the pre-configured unit test suite (includes UC-02).
4. System collects the test results (pass/fail per test case, coverage, syntax check output).
5. System applies the assignment's rubric weighting rules to the test results.
6. System generates an itemized rubric score breakdown (per-criterion scores and a total).
7. System stores the report and notifies the student and faculty evaluator that scoring is complete.
8. Use case ends successfully.

Alternate Flow — A1: Low-Confidence Score (extends to UC-06)

Trigger: at Step 6, the computed rubric score falls within the configured low-confidence variance threshold.

1. System flags the submission and adds it to the faculty "Needs Review" queue (UC-06 — extend).
2. Faculty Evaluator manually inspects the test output and rubric breakdown.
3. Faculty Evaluator confirms or overrides the automated score (UC-03 / FR-003).
4. Flow resumes at Step 7 of the main success scenario with the faculty-confirmed score.

Alternate Flow — A2: Invalid or Corrupted Submission File

Trigger: at Step 2, the uploaded file cannot be unzipped or is missing required project structure.

1. System rejects the file and marks the submission status as "Failed — Invalid Format."
2. System notifies the student with the specific validation error.
3. Student corrects the submission and re-uploads, returning to Step 1.